TraceBack — Costing

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Running TraceBack costs about US$24.84 (₹2,360) a month at AWS on-demand prices, and the team pays nothing: the AWS Free plan credits cover the server, and every other service is on a free tier. The single biggest cost is the EC2 instance (78%). A one-year savings plan would cut the bill to about US$17.69 a month if the portal were kept beyond the course.

TraceBack is the IIT Kanpur Lost & Found portal. Prices are AWS on-demand list prices for the Sydney region (`ap-southeast-2`), where the server runs, in US dollars, with rupees at ₹95 = US$1 (early November 2026). A month is 730 hours.

1. Monthly running cost

Item	What we use	Unit price	Per month (US$)	Per month (₹)
EC2 instance	1 × `t3.small` (2 vCPU, 2 GB), Linux, always on	$0.0264 per hour	19.27	1,831
Disk	20 GB EBS gp3	$0.096 per GB-month	1.92	182
Public IPv4 address	1 Elastic IP	$0.005 per hour	3.65	347
Data transfer out	about 0.8 MB per page view (after thumbnails)	first 100 GB a month free	0.00	0
HTTPS certificate	Let’s Encrypt through Caddy	free	0.00	0
Total			**24.84**	**2,360**

The 100 GB of free data transfer covers about 125,000 page views a month, far above campus use: at 2,000 page views a day TraceBack would send about 48 GB.

2. What the project actually pays

- **US$0.** The account is on the AWS Free plan, which gives US$100 in credits at sign-up and up to US$100 more for completing short console activities, for up to 6 months. At US$24.84 a month, US$100 lasts four months (November 2026 to February 2027) and US$200 the full six.
- **CPU credits:** a `t3.small` earns 20% of each core as its baseline. After the performance work a page view needs about a third of the CPU it did, so normal campus use stays inside the baseline and no surplus-credit charges ($0.05 per vCPU-hour) are expected.
- **When the course ends:** terminate the instance and release the Elastic IP, or both keep charging.

3. Ways to pay less

Option	Per month (US$)	Saving	Trade-off
Current: on-demand, always on	24.84	–	none
Stop the server between demos	5.57 (disk and IP only)	78%	Site offline while stopped; restart takes about 2 minutes
1-year EC2 Instance Savings Plan or Reserved Instance ($0.0166 per hour)	17.69	29%	One-year commitment; worth it only if the portal stays in use
`t3.micro` (1 GB)	–	–	Not viable: the Next.js build and the AI models need 2 GB plus swap
Spot instance	–	–	Not suitable: AWS can stop it at any time, and the portal must stay up

4. Outside services

Service	Use	Plan	Cost
Groq (`openai/gpt-oss-20b`)	Main model for the AI matching agent	Free tier	₹0

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Google Gemini (`gemini-2.5-flash`)	Fallback model and photo descriptions	Free tier	₹0
Gmail SMTP	Sign-in codes and notices	Free account with an app password	₹0
sslip.io	Domain name for the server's IP	Free	₹0
GitHub	Repository, pull requests, reviews	Free organisation	₹0
Jira	Backlog and sprints	Free plan (the site was suspended after the team passed its member limit)	₹0

The AI cost is kept at zero by design: the agent is called only when the maths is unsure (`AGENT_MATCHING=ambiguous`), and the app never makes more than 500 AI calls a day (`AI_DAILY_CALL_LIMIT`). The free tiers rate-limit rather than bill; a Groq rate-limit reply (429) was seen once, and the app then used Gemini. If the same limit were used every day on paid plans, it would cost about US$4.70 a month on Groq ($0.075 per million input tokens, $0.30 per million output, about 3,000 tokens in and 300 out per call) plus about US$3 for 100 Gemini photo descriptions a day ($0.30 and $2.50 per million tokens).

5. Development effort

For comparison, a classic size-based estimate. TraceBack's product code is about 13,300 non-blank, non-comment lines: backend 5,760, frontend 6,650, database migrations 640, deployment scripts 220; the 6,200 lines of tests and test scripts are not counted. Basic COCOMO for an organic project gives:

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E = 2.4 \times 13.3^{1.05} \approx 36\ \text{person-months}, \qquad T = 2.5 \times E^{0.38} \approx 9.8\ \text{months}
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At a notional ₹50,000 per person-month for an entry-level developer in India, that is about ₹18.1 lakh (US$19,100). The team of six students delivered it part-time between 19 Aug and 6 Nov 2026 (11 weeks), using an AI coding assistant throughout (see the AI Engineering Log), so the estimate mostly shows how much the AI assistance and the reuse of open-source libraries saved.

Sources

- AWS EC2 t3.small price in ap-southeast-2 and regional price list (EBS gp3 $0.096 per GB-month), checked 6 Nov 2026
- Amazon EC2 T3 instances (20% baseline, $0.05 per vCPU-hour surplus credits)
- AWS Free Tier data transfer: 100 GB a month
- Groq gpt-oss-20b prices and Gemini 2.5 Flash prices
- Exchange rate: cross-rate from the Nepal Rastra Bank figures, 2 Nov 2026